

Dimitris Sacharidis

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Computer & Decision Engineering Department
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Personal Information

Date of Birth March 26, 1979
Nationality Greek
Marital Status Married (3 children)
Languages Greek (native), English (C2), German (B2)

Positions

Jan '21 – **Assistant Professor of Data Science and Data Engineering**, *Web & Information Technologies Lab, Computer & Decision Engineering Department, Université Libre de Bruxelles, Belgium*
Feb '15 – **University Assistant**, *E-Commerce Research Unit, Institute of Information Systems Engineering, Technische Universität Wien, Austria*
Jan '21
Mar '09 – **Postdoctoral Researcher**, *Institute for the Management of Information Systems, “Athena” Research Center, Greece*
Jan '15
Mar '09 – **Postdoctoral Researcher**, *University of Science and Technology, Hong Kong, China*
Feb '11 (*secondment*)
Oct '08 – **Visiting Lecturer**, *University of Peloponnese, Greece*
Feb '09

Research Interests

Big Data Management

Rich (Spatial-Temporal-Textual-Social) Data Mining; Data Summarization; Indexing and Data Structures

Data Science

(Social, Group, Location-Aware) Recommender Systems; Ethics in Data Science

Education

Oct '04 – **Ph.D., Computer Science**, *National Technical University of Athens, Greece*
Feb '09 Thesis — “*Data Streams Management*” awarded at 13.02.2009
Advisor — *Prof. Timos Sellis*
Aug '02 – **Master of Science, Computer Science**, *University of Southern California, U.S.A*
May '04 Advisor — *Prof. Cyrus Shahabi*

Oct '96 – **Diploma in Electrical and Computer Engineering**, *National Technical University of Athens*, Greece
 Oct '01 Thesis — “*Mining Fuzzy Association Rules*”
 Advisor — *Prof. Timos Sellis*

Third Party Funding

May '20 – *A Fair Multi-stakeholder Job Market Recommender Leveraging Personality*,
 Nov '20' in collaboration with mediumrare GmbH Austria (5 KEur) — *PI*
 Oct '18 – *Data Management and Analytics of User Mobility Traces*,
 Mar '19 in collaboration with Flink GmbH Austria, funded by Austrian Research Promotion Agency (FFG) Innovationsscheck (12 KEur) — *PI*
 Aug '18 – PredOpi — *Modern Machine Learning Techniques for Opinion Prediction from Past User Interactions*,
 Dec '18 in collaboration with eMentalist GmbH Austria (6 KEur) — *PI*
 Feb '18 – *Recommender Systems and Chatbots for Hotels*,
 Jul '18 in collaboration with UpSell Inc. Greece, funded by DataPitch Innovation Program (100 KEur) — *work package leader*
 Mar '13 – EICOS — *Foundations for Personalized Cooperative Information Systems*,
 Jul '15 “Thalis” call (operation program “Education and Lifelong Learning”, NSRF 2007–2013) of Ministry of Education, Greece — *work package leader*
 Mar '09 – BIOSEC — *Security and Privacy in Life Sciences Data*,
 Feb '12 FP7-PEOPLE-2008-IOF project (214 KEur) — *PI*
 Jan '05 – PENED '03 — *Data Management for Location-based Services*,
 Dec '07 funded by 75% E.U., 25% General Secretariat of Research and Technology, Greece — *co-PI*
 May '03 – ProDA — *An End-to-End Wavelet-Based Progressive Data Analysis System*,
 Jul '04 funded by U.S.A. NSF grants EEC-9529152 (IMSC ERC) and IIS-0238560 (PECASE) — *work package leader*

List of Publications

ORCID <http://orcid.org/0000-0001-5022-1483>

DBLP <http://dblp.uni-trier.de/pers/hd/s/Sacharidis:Dimitris>

Google <https://scholar.google.com/citations?user=0cSioFgAAAAJ&hl=en>

Scholar (citations: 1336, h-index: 20, i10-index: 30)

Top Publications (A* and A)	Conference Rank Publications			Journal Rank Publications		
	SIGMOD	A*	[C1]	VLDBJ	A*	[J7, J3, J2]
	SIGKDD	A*	[C5]	TKDE	A	[J13, J6, J5]
	ICDE	A*	[C60, C14, C12]			
	EDBT	A	[C37, C36, C32, C23, C17, C11, C7, C3]			
	SIGSPATIAL	A	[C44, C41, C40, C39, C35, C29, C26, C25, C20]			

(ranks according to CORE: <http://www.core.edu.au/conference-portal>)

Book Chapters

- [BC2] G. Gratsias, D. Sacharidis, M. Terrovitis, and T. Sellis. “Privacy in the publication and exchange of data”. In: *Privacy in Information and Communication Technologies: Technical and Legal Issues*. Papasotiriou, 2010, pp. 469–501.
- [BC1] C. Shahabi, D. Sacharidis, and M. Jahangiri. “Wavelets for Querying Multidimensional Datasets”. In: *Encyclopedia of Data Warehousing and Mining*. BC1. Idea Group Reference, 2005, pp. 1196–1200.

Journal Articles

- [J13] I. Kanellos, T. Vergoulis, D. Sacharidis, T. Dalamagas, and Y. Vassiliou. “Impact-Based Ranking of Scientific Publications: A Survey and Experimental Evaluation”. In: *IEEE Transactions on Knowledge and Data Engineering (TKDE)* (2019).
- [J12] C. P. Mukamakuza, D. Sacharidis, and H. Werthner. “The Role of Activity and Similarity in Rating and Social Behavior in Social Recommender Systems”. In: *International Journal on Artificial Intelligence Tools* (2019).
- [J11] K. Benouaret, D. Sacharidis, D. Benslimane, and A. Hadjali. “Selecting Services for Multiple Users: Let’s Be Democratic”. In: *IEEE Transactions on Services Computing (TSC)* (2018).
- [J10] S. Qi, D. Sacharidis, P. Bouros, and N. Mamoulis. “Snapshot and continuous points-based trajectory search”. In: *International Journal on Advances of Computer Science for Geographic Information Systems (GeoInformatica)* (2016).
- [J9] N. Bikakis, K. Benouaret, and D. Sacharidis. “Finding desirable objects under group categorical preferences”. In: *Knowledge and Information Systems (KAIS)* 49.1 (2016), pp. 273–313.
- [J8] G. Tsatsanifos, D. Sacharidis, and T. Sellis. “Index-based Query Processing on Distributed Multidimensional Data”. In: *International Journal on Advances of Computer Science for Geographic Information Systems (GeoInformatica)* 17.3 (2013), pp. 489–519.
- [J7] T. Vergoulis, T. Dalamagas, D. Sacharidis, and T. Sellis. “Approximate Regional Sequence Matching for Genomic Databases”. In: *The International Journal on Very Large Data Bases (VLDBJ)* 21.6 (2012), pp. 779–795.
- [J6] P. Bouros, D. Sacharidis, T. Dalamagas, S. Skiadopoulos, and T. Sellis. “Evaluating Path Queries over Frequently Updated Route Collections”. In: *IEEE Transactions on Knowledge and Data Engineering (TKDE)* 24.7 (2012), pp. 1276–1290.
- [J5] D. Skoutas, D. Sacharidis, A. Simitsis, and T. K. Sellis. “Ranking and Clustering Web Services using Multi-Criteria Dominance Relationships”. In: *IEEE Transactions on Services Computing (TSC)* 3.3 (2010), pp. 163–177.
- [J4] D. Sacharidis, K. Mouratidis, and D. Papadias. “k-Anonymity in the Presence of External Databases”. In: *IEEE Transactions on Knowledge and Data Engineering (TKDE)* 2.3 (2010), pp. 392–403.
- [J3] K. Mouratidis, D. Sacharidis, and H.-H. Pang. “Partially Materialized Digest Scheme: An Efficient Verification Method for Outsourced Databases”. In: *The International Journal on Very Large Data Bases (VLDBJ)* 18.1 (2009), pp. 363–381.
- [J2] D. Sacharidis, A. Deligiannakis, and T. Sellis. “Hierarchically Compressed Wavelet Synopses”. In: *The International Journal on Very Large Data Bases (VLDBJ)* 18.1 (2009), pp. 203–231.
- [J1] C. Shahabi, M. Jahangiri, and D. Sacharidis. “Hybrid Query and Data Ordering for Fast and Progressive Range-Aggregate Query Answering”. In: *International Journal of Data Warehousing and Mining* 1.2 (2005). J1, pp. 49–69.

Conference Papers

- [C60] I. Kanellos, T. Vergoulis, D. Sacharidis, T. Dalamagas, and Y. Vassiliou. “Ranking Papers by their Short-Term Scientific Impact”. In: *IEEE International Conference on Data Engineering (ICDE)*. IEEE, 2021, pp. 1997–2002.
- [C59] G. Giannopoulos, G. Papastefanatos, D. Sacharidis, and K. Stefanidis. “Interactivity, Fairness and Explanations in Recommendations”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP)*. Ed. by J. Masthoff, E. Herder, N. Tintarev, and M. Tkalcic. ACM, 2021, pp. 157–161.
- [C58] V. Kaffes, D. Sacharidis, and G. Giannopoulos. “Model-Agnostic Counterfactual Explanations of Recommendations”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP)*. Ed. by J. Masthoff, E. Herder, N. Tintarev, and M. Tkalcic. ACM, 2021, pp. 280–285.
- [C57] D. Sacharidis. “Building User Trust in Recommendations via Fairness and Explanations”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP) Adjunct*. 2020.
- [C56] D. Defiebre, P. Germanakos, and D. Sacharidis. “DANOS: A Human-Centered Decentralized Simulator in SIOT”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP) Adjunct*. 2020.
- [C55] D. Defiebre, D. Sacharidis, and P. Germanakos. “A Decentralized Recommendation Engine in the Social Internet of Things”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP) Adjunct*. 2020.
- [C54] D. Sacharidis, C. P. Mukamakuza, and H. Werthner. “Fairness and Diversity in Social-Based Recommender Systems”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP) Adjunct*. 2020.
- [C53] J. Luef, C. Ohrfandl, D. Sacharidis, and H. Werthner. “A Recommender System For Investing In Early-Stage Enterprises”. In: *ACM/SIGAPP Symposium On Applied Computing (SAC)*. 2020.
- [C52] M. Ion, D. Sacharidis, and H. Werthner. “Designing a Recommender System for Board Games”. In: *ACM/SIGAPP Symposium On Applied Computing (SAC)*. 2020.
- [C51] L. Bellatreche, F. Bentayeb, M. Bieliková, O. Boussaid, B. Catania, P. Ceravolo, E. Demidova, M. H. Ferrari, M. T. G. López, C. S. Hara, S. Kordic, I. Lukovic, A. Mannocci, P. Manghi, F. Osborne, C. Papatheodorou, S. Ristic, D. Sacharidis, O. Romero, A. A. Salatino, G. Talens, M. van Keulen, T. Vergoulis, and M. Zumer. “Databases and Information Systems in the AI Era: Contributions from ADBIS, TPD L and EDA 2020 Workshops and Doctoral Consortium”. In: *ADBIS/TPDL/EDA Workshops*. Vol. 1260. Communications in Computer and Information Science. Springer, 2020, pp. 3–20.
- [C50] L. Weydemann, D. Sacharidis, and H. Werthner. “Defining and Measuring Fairness in Location Recommendations”. In: *ACM SIGSPATIAL Workshop on Location-Based Recommendations, Geosocial Networks, and Geoadvertising (LocalRec)*. 2019.
- [C49] D. Sacharidis, K. Mouratidis, and D. Kleftogiannis. “A Unified Approach for Consumer and Provider Fairness”. In: *ACM Conference on Recommender Systems (RecSys) LBR*. 2019.
- [C48] C. P. Mukamakuza, D. Sacharidis, and H. Werthner. “The Impact of Social Connections in Personalization”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP) Adjunct*. 2019.
- [C47] D. Sacharidis. “Diversity and Novelty in Social-Based Collaborative Filtering”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP)*. 2019.

- [C46] D. Sacharidis. “Modeling Uncertainty in Group Recommendations”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP)*. 2019.
- [C45] D. Sacharidis. “Top-N Group Recommendations with Fairness”. In: *ACM/SIGAPP Symposium On Applied Computing (SAC)*. 2019.
- [C44] D. Skoutas, D. Sacharidis, and K. Patroumpas. “Efficient Progressive and Diversified Top-k Best Region Search”. In: *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. 2018.
- [C43] D. Sacharidis, P. Mehta, D. Skoutas, K. Patroumpas, and A. Voisard. “Selecting Representative and Diverse Spatio-Textual Posts over Sliding Windows”. In: *International Conference on Scientific and Statistical Database Management (SSDBM)*. 2018.
- [C42] C. P. Mukamakuza, D. Sacharidis, and H. Werthner. “Mining User Behavior in Social Recommender Systems”. In: *International Conference on Web Intelligence, Mining and Semantic (WIMS)*. 2018.
- [C41] D. Sacharidis, P. Bouros, and T. Chondrogiannis. “Finding The Most Preferred Path”. In: *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. 2017.
- [C40] Q. Gu, D. Sacharidis, M. Mathioudakis, and G. Wang. “Inferring Venue Visits from GPS Trajectories”. In: *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. 2017.
- [C39] D. Sacharidis, P. Mehta, D. Skoutas, K. Patroumpas, and A. Voisard. “Continuous Summarization of Streaming Spatio-Textual Posts”. In: *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. 2017.
- [C38] D. Sacharidis. “Group Recommendations by Learning Rating Behavior”. In: *ACM Conference on User Modeling, Adaptation and Personalization (UMAP)*. ACM, 2017, pp. 174–182.
- [C37] P. Mehta, M. Kotlarski, D. Skoutas, D. Sacharidis, K. Patroumpas, and A. Voisard. “ μ TOP: Spatio-Temporal Detection and Summarization of Locally Trending Topics in Microblog Posts (demo)”. In: *International Conference on Extending Database Technology (EDBT)*. 2017.
- [C36] P. Mehta, D. Sacharidis, D. Skoutas, and A. Voisard. “Finding Socio-Textual Associations Among Locations”. In: *International Conference on Extending Database Technology (EDBT)*. 2017.
- [C35] P. Mehta, D. Skoutas, D. Sacharidis, and A. Voisard. “Coverage and Diversity Aware Top-k Query for Spatio-Temporal Posts”. In: *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. 2016.
- [C34] A. Papatriantafyllou and D. Sacharidis. “High Performance Parallel Summed-Area Table Kernels for Multi-core and Many-core Systems”. In: *International Conference on Parallel and Distributed Computing (Euro-Par)*. Vol. 9833. Lecture Notes in Computer Science. Springer, 2016, pp. 306–318.
- [C33] P. Mehta, D. Sacharidis, D. Skoutas, and A. Voisard. “Keyword-Based Retrieval of Frequent Location Sets in Geotagged Photo Trails”. In: *ACM Web Science Conference (WebSci)*. 2016.
- [C32] D. Skoutas, D. Sacharidis, and K. Stamatoukos. “Identifying and Describing Streets of Interest”. In: *International Conference on Extending Database Technology (EDBT)*. 2016.
- [C31] D. Sacharidis, A. Delic, and J. Neidhardt. “Learning the Role and Behavior of Users in Group Decision Making”. In: *International Workshop on Decision Making and Recommender Systems (DMRS)*. 2015.

- [C30] P. Bouros, N. Lathia, M. Renz, F. Ricci, and D. Sacharidis. “LocalRec’15: Workshop on Location-Aware Recommendations”. In: *ACM Conference on Recommender Systems (RecSys)*. 2015, pp. 351–352.
- [C29] D. Sacharidis and A. Deligiannakis. “Spatial Cohesion Queries”. In: *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. 2015.
- [C28] S. Qi, P. Bouros, D. Sacharidis, and N. Mamoulis. “Efficient Point-based Trajectory Search”. In: *International Symposium on Spatial and Temporal Databases (SSTD)*. 2015.
- [C27] D. Sacharidis and T. Sellis. “Efficient Identification of the Highest Diversity Gain Object”. In: *International Workshop on Information Search, Integration and Personalization (ISIP) (invited)*. 2014.
- [C26] D. Sacharidis, D. Skoutas, and G. Skoumas. “Continuous Monitoring of Nearest Trajectories”. In: *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. 2014.
- [C25] P. Bouros, D. Sacharidis, and N. Bikakis. “Regionally Influential Users in Location-Aware Social Networks”. In: *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. 2014.
- [C24] N. Bikakis, D. Sacharidis, and T. Sellis. “A Study on External Memory Scan-based Skyline Algorithms”. In: *International Conference on Database and Expert Systems Applications (DEXA)*. 2014.
- [C23] G. Tsatsanifos, D. Sacharidis, and T. Sellis. “RIPPLE: A Scalable Framework for Distributed Processing of Rank Queries”. In: *International Conference on Extending Database Technology (EDBT)*. 2014.
- [C22] N. Bikakis, K. Benouaret, and D. Sacharidis. “Reconciling Multiple Categorical Preferences with Double Pareto-based Aggregation”. In: *International Conference on Database Systems for Advanced Applications (DASFAA)*. 2014.
- [C21] I. Kanellos, T. Vergoulis, D. Sacharidis, T. Dalamagas, A. Hatzigeorgiou, S. Sartzetakis, and T. Sellis. “MR-microT: A MapReduce-based MicroRNA Target Prediction Method (demo)”. In: *International Conference on Scientific and Statistical Database Management (SSDBM)*. 2014.
- [C20] D. Sacharidis and P. Bouros. “Routing Directions: Keeping it Fast and Simple”. In: *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. 2013.
- [C19] K. Benouaret, D. Sacharidis, D. Benslimane, and A. Hadjali. “Majority-Rule-Based Web Service Selection”. In: *International Conference on Web Information System Engineering (WISE)*. 2012.
- [C18] S. Auer, T. Dalamagas, H. E. Parkinson, F. Bancilhon, G. Flouris, D. Sacharidis, P. Buneman, D. Kotzinos, Y. Stavrakas, V. Christophides, G. Papastefanatos, and K. Thiveos. “Diachronic linked data: towards long-term preservation of structured interrelated information”. In: *International Workshop on Open Data (WOD)*. 2012, pp. 31–39.
- [C17] G. Tsatsanifos, D. Sacharidis, and T. Sellis. “On enhancing scalability for distributed RDF/S stores”. In: *International Conference on Extending Database Technology (EDBT)*. 2011.
- [C16] G. Tsatsanifos, D. Sacharidis, and T. Sellis. “MIDAS: Multi-attribute Indexing for Distributed Architecture Systems”. In: *International Symposium on Spatial and Temporal Databases (SSTD)*. 2011.

- [C15] P. Bouros, D. Sacharidis, T. Dalamagas, and T. Sellis. “Dynamic Pickup and Delivery with Transfers”. In: *International Symposium on Spatial and Temporal Databases (SSTD)*. 2011.
- [C14] D. Sacharidis, A. Arvanitis, and T. Sellis. “Probabilistic Contextual Skylines”. In: *IEEE International Conference on Data Engineering (ICDE)*. 2010.
- [C13] P. Bouros, S. Skiadopoulos, T. Dalamagas, D. Sacharidis, and T. Sellis. “Evaluating reachability queries over path collections”. In: *International Conference on Scientific and Statistical Database Management (SSDBM)*. C11. Springer, 2008, pp. 455–472.
- [C12] D. Sacharidis, S. Papadopoulos, and D. Papadias. “Topologically Sorted Skylines for Partially Ordered Domains”. In: *IEEE International Conference on Data Engineering (ICDE)*. 2009, pp. 1072–1083.
- [C11] D. Skoutas, D. Sacharidis, A. Simitsis, V. Kantere, and T. K. Sellis. “Top-k dominant web services under multi-criteria matching”. In: *International Conference on Extending Database Technology (EDBT)*. 2009, pp. 898–909.
- [C10] D. Skoutas, D. Sacharidis, V. Kantere, and T. Sellis. “Efficient Semantic Web Service Discovery in Centralized and P2P Environments”. In: *International Semantic Web Conference (ISWC)*. C10. Springer, 2008, pp. 583–598.
- [C9] D. Skoutas, D. Sacharidis, A. Simitsis, and T. Sellis. “Serving the Sky: Discovering and Selecting Semantic Web Services through Dynamic Skyline Queries”. In: *IEEE International Conference on Semantic Computing (ICSC)*. C9. IEEE Computer Society, 2008.
- [C8] D. Sacharidis, P. Bouros, and T. Sellis. “Caching Dynamic Skyline Queries”. In: *International Conference on Scientific and Statistical Database Management (SSDBM)*. Vol. 5069. Lecture Notes in Computer Science. C8. Springer, 2008, pp. 455–472.
- [C7] D. Sacharidis, K. Patroumpas, M. Terrovitis, V. Kantere, M. Potamias, K. Mouratidis, and T. Sellis. “On-line Discovery of Hot Motion Paths”. In: *International Conference on Extending Database Technology (EDBT)*. Vol. 261. ACM International Conference Proceeding Series. C10. ACM, 2008, pp. 392–403.
- [C6] S. Papadopoulos, D. Sacharidis, and K. Mouratidis. “Continuous Medoid Queries over Moving Objects”. In: *International Symposium on Spatial and Temporal Databases (SSTD)*. C6. Springer, 2007.
- [C5] P. Karras, D. Sacharidis, and N. Mamoulis. “Exploiting Duality in Summarization with Deterministic Guarantees”. In: *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. C5. ACM, 2007, pp. 380–389.
- [C4] M. Mathioudakis, D. Sacharidis, and T. Sellis. “A study on Workload-aware Wavelet Synopses for Point and Range-sum Queries”. In: *ACM International Workshop on Data Warehousing and OLAP (DOLAP)*. C4. ACM, 2006.
- [C3] G. Cormode, M. Garofalakis, and D. Sacharidis. “Fast Approximate Wavelet Tracking on Streams”. In: *International Conference on Extending Database Technology (EDBT)*. C3. Springer, 2006, pp. 4–22.
- [C2] D. Sacharidis. “Constructing Optimal Wavelet Synopses”. In: *International Conference on Extending Database Technology (EDBT) Workshops*. C2. Springer, 2006, pp. 97–104.
- [C1] M. Jahangiri, D. Sacharidis, and C. Shahabi. “SHIFT-SPLIT: I/O Efficient Maintenance of Wavelet-Transformed Multidimensional Data”. In: *ACM SIGMOD International Conference on Management of Data (SIGMOD)*. C1. 2005, pp. 275–286.